Explainable Anomaly Detection in Hospital Supply Chains

Abstract. Hospital supply chains in tier two Indian cities face stock-outs of essential drugs that are hard to predict. We combine isolation forests with SHAP explanations so pharmacists can see which suppliers and seasons drive each alert.

Twelve months of procurement records from three district hospitals were cleaned and aligned to a common drug formulary before modelling.

Pharmacists rated eighty-one percent of the explanations as useful, and the alert precision at the top fifty anomalies reached zero point seven four.